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Concordia University
Faculty of Engineering and Computer Science
Applied Ordinary Differential Equations - ENGR213/4
Winter 2023 Final Examination

Instructors: Korotkin D, Paradis, A., Shnirelman, A.

Special instructions: Do all problems. Each problem worth 10 marks. Show all steps or else no marks will be given. Only Faculty approved calculators are allowed. No other aids are allowed.

1. Find the general solution of the following equation by separation of variables:
$$e^x y \frac{dy}{dx} = e^{-2y} + e^{-3x-2y}$$

2. Find the general solution of the linear differential equation:
$$y \, dx - 4(x + y^8) \, dy = 0$$

3. Find the general solution of the homogeneous differential equation:
$$(y^2 + xy) \, dx - x^2 \, dy = 0$$

4. Solve the initial-value problem for the following Bernoulli equation:
$$x^2 \frac{dy}{dx} - 2xy = 3y^6 \quad y(1) = 3$$

5. Determine whether the following differential equation is exact:
$$(5x^4y - 15x^2 + 8xy) \, dx + (x^5 + 4x^2 - 1) \, dy = 0$$

If yes, find the implicit solution of the initial value problem $y(0) = 1$.

6. Using the method of undetermined coefficients find the general solution of the differential equation
$$y'' - 4y' + 4y = (x - 2)e^{2x}$$

7. Solve the following initial value problem by the method of your choice:
$$x^2 y'' - 4xy' + 6y = 3x^{-3} \quad y(1) = 0, \quad y'(1) = 1$$

1

$$\Delta (y^2 + xy) dx - x^2 dy = 0$$

homogeneous $\Rightarrow y^2(1 + \frac{x}{y}) dx - y^2(\frac{x}{y})^2 dy = 0$

$$u = \frac{x}{y} \Rightarrow x = uy, dx = u dy + y du$$

$$\Rightarrow y^2(1 + u)(u dy + y du) - y^2 u^2 dy = 0$$

$$\frac{(y^2 + uy^2)(u dy + y du)}{(y^2 + uy^2)(u dy + y du)} - y^2 u^2 dy = 0$$

$$y^2 u dy + y^3 du + u^2 y^2 dy + u y^3 du - y^2 u^2 dy = 0$$

$$y^2 u dy = -y^3(1 + u) du$$

$$\frac{dy}{y} = -\frac{1}{u} du - du \quad \text{integrating}$$

$$\Rightarrow \ln|y| = -\ln|u| - u + C_1$$

$$-\ln|y| = \ln|u| + u + C$$

$$\ln\left|\frac{1}{y}\right| = \ln\left|\frac{x}{y}\right| + \frac{x}{y} + C$$

[#4] $x^2 \frac{dy}{dx} - 2xy = 3y^6, y(1) = 3$

$$\frac{dy}{dx} - \frac{2y}{x} = \frac{3}{x^2} y^6 \Rightarrow n = 6 \therefore u = y^{1-6} = y^{-5}$$

$$\frac{dy}{dx} = \frac{y^6}{(-5)} \frac{du}{dx} \Rightarrow \frac{y^6}{(-5)} \frac{du}{dx} + \frac{2}{x} y = \frac{3}{x^2} y^6 \div y^6$$

$$-\frac{1}{5} \frac{du}{dx} - \frac{2}{x} u = \frac{3}{x^2} \times (-5) \Rightarrow \left[\frac{du}{dx} + \frac{10}{x} u = -\frac{15}{x^2} \right]$$

$-9/17$

$(k_{21}) = 0 \quad \text{and} \quad 4k_{11} - 2k_{21} = 0 \rightarrow 4k_{11} = 0 \rightarrow k_{11} = 0$
 $\lambda_2 = -1$
 $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$ ✓

$\begin{pmatrix} 2 & -1 \\ -4 & 2 \end{pmatrix} \begin{pmatrix} k_{12} \\ k_{22} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \rightarrow 2k_{12} = k_{22}$
 $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ ✓
 $\therefore X_p = C_1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{3t} + C_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{-t}$ ✓

$X(t) = C_1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{3t} + C_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{-t} + \begin{pmatrix} t - 5/3 + 1/4 e^t \\ 4t - 8/3 \end{pmatrix}$ ✓

✓

$$\Rightarrow \frac{1}{3^5} = -\frac{15}{9}x + Cx^{-10}$$

$$u = -\frac{15}{9}x + Cx^{-10}$$

$$\text{IUP: } y(0)=3 \Rightarrow \frac{1}{3^5} = -\frac{15}{9} + C \rightarrow C = \frac{1}{3^5} + \frac{15}{9} = 2\frac{43}{9}$$

$$\therefore y^5 = \frac{1}{-\frac{15}{9}x + \frac{(243+15)}{9}x^{-10}}$$

(#5) exact. $\underbrace{(5x^4y - 15x^2 + 8xy)}_M dx + \underbrace{(x^5 + 4x^2 - 1)}_N dy = 0, y(0)=1$

$$\frac{\partial M}{\partial y} = 5x^4 + 8x \quad \frac{\partial N}{\partial x} = 5x^4 + 8x \quad \text{or. } \boxed{3} \quad (w)$$

$$\therefore f(x,y) = \int (5x^4y - 15x^2 + 8xy) dx + g(y) = x^5y - 5x^3 + 4x^2y + g(y)$$

$$\frac{\partial f(x,y)}{\partial y} = x^5 + 4x^2 - 1 = x^5 + 4x^2 + g'(y) \Rightarrow g'(y) = -1 \quad g(y) = -y \quad (y)$$

$$\therefore f(x,y) = C \Rightarrow \boxed{x^5y - 5x^3 + 4x^2y - y = C} \quad y(0)=1$$

$$-1 = C$$

$$\boxed{x^5y - 5x^3 + 4x^2y - y = -1}$$

$$x^2 y'' - 4xy' + 6y = 3x^3, \quad y(1) = 0, \quad y'(1) = 1.$$

Ans: $x^m \Rightarrow m(m-1) - 4m + 6 = 0 \Rightarrow m = 5 \pm \sqrt{25-24} = \begin{cases} 3 \\ 2 \end{cases}$

$$\therefore y_c(x) = C_1 x^3 + C_2 x^2$$

$$\text{NH} \rightarrow y'' - \frac{4}{x} y' + \frac{6}{x^2} y = 3x^{-5} \quad y_p = u y_1 + v y_2$$

$$\rightarrow \text{let } y_1 = x^3; y_2 = x^2; f(x) = 3x^{-5}$$

$$W = \begin{vmatrix} x^3 & x^2 \\ 3x^2 & 2x \end{vmatrix} = 2x^4 - 3x^4 = -x^4$$

$$\rightarrow u = \int \frac{-y_2 f(x) dx}{W} = \int \frac{x^2 \cdot 3x^{-5} dx}{-x^4} = -3 \int x^{-3} dx = -2x^{-2}$$

$$v = \int \frac{y_1 f(x) dx}{W} = \int \frac{x^3 \cdot 3x^{-5} dx}{-x^4} = -\frac{3}{5} x^{-2}$$

$$\therefore y_p = -2x^{-2} + \left(-\frac{3}{5}\right)x^{-2} = -\frac{13}{5}x^{-2}$$

$$\therefore y(x) = C_1 x^3 + C_2 x^2 - \frac{13}{5} x^{-2}$$

$$y(1) = 0 = C_1 + C_2 - \frac{13}{5} \Rightarrow C_1 = \frac{13}{5} - C_2$$

$$y'(x) = 3C_1 x^2 + 2C_2 x - \frac{26}{5} x^{-3} \quad 1 = -C_2 \Rightarrow C_2 = -1$$

$$y'(1) = 1 = 3C_1 + 2C_2 - \frac{26}{5}$$

$$y(x) = \frac{18}{5} x^3 - x^2 - \frac{13}{5} x^{-2}$$

$$\frac{dx}{dt} = x - y + 3t$$

$$\frac{dy}{dt} = y - 4x + e^t$$

$$\Rightarrow \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 3t \\ e^t \end{pmatrix}$$

$$\begin{vmatrix} 1-\lambda & -1 \\ -4 & 1-\lambda \end{vmatrix} = 0 \Rightarrow \begin{cases} 1-2\lambda+\lambda^2-4=0 \\ \lambda^2-2\lambda-3=0 \end{cases} \Rightarrow \lambda_{1,2} = \frac{2 \pm \sqrt{4+12}}{2} = \begin{cases} 3 \\ -1 \end{cases}$$

Let us use $\lambda = 1 + \sqrt{2}i$

$$\Rightarrow \begin{pmatrix} 1-1-\sqrt{2}i & -1 \\ -4 & 1-1-\sqrt{2}i \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} -\sqrt{2}i k_1 - k_2 = 0 \\ -4k_1 - \sqrt{2}i k_2 = 0 \end{cases}$$

$$\therefore k = \begin{pmatrix} 1 \\ \sqrt{2}i \end{pmatrix} \Rightarrow B_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad B_2 = \begin{pmatrix} 0 \\ \sqrt{2} \end{pmatrix}$$

$$\Rightarrow X_c(t) = C_1 \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cos(\sqrt{2}t) - \begin{pmatrix} 0 \\ \sqrt{2} \end{pmatrix} \sin(\sqrt{2}t) \right] e^t + C_2 \left[\begin{pmatrix} 0 \\ \sqrt{2} \end{pmatrix} \cos(\sqrt{2}t) + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \sin(\sqrt{2}t) \right] e^t$$

② $X_p \sim \begin{pmatrix} At+B+Ce^t \\ Dt+E+Fe^t \end{pmatrix} \quad \dot{X}_p = \begin{pmatrix} A+Ce^t \\ D+Fe^t \end{pmatrix}$

$$\therefore \begin{pmatrix} A+Ce^t \\ D+Fe^t \end{pmatrix} = \begin{pmatrix} A+B+Ce^t-Dt-E-Fe^t+3t \\ -4A-4B-4Ce^t+Dt+E+Fe^t+e^t \end{pmatrix}$$

① $t^0 \mid A=B-E \quad 1+E=B$
 $t^1 \mid 0=A+B+3 \Rightarrow A+B=-3$
 $t^e \mid C=C-F \Rightarrow 0=-F$

② $t^0 \mid D=-4B+E \Rightarrow 4=-4+E \Rightarrow E=8$
 $t^1 \mid 0=-4A+D \Rightarrow D=4A$
 $t^e \mid F=-4C+F+1 \Rightarrow 4C=1 \Rightarrow C=\frac{1}{4}$

$$\Rightarrow X_p = \begin{pmatrix} t - \frac{5}{3} + \frac{1}{4}e^t \\ 4t - \frac{5}{3} \end{pmatrix}$$

$$\boxed{1} \quad e^x y \frac{dy}{dx} = e^{-2y} + e^{-3x-2y} = e^{-2y} (1 + e^{-3x}) \quad] / 3$$

$$y e^{2y} dy = (e^{-x} + e^{-4x}) dx$$

$$\int y e^{2y} dy = \int e^{-x} dx + \int e^{-4x} dx + C \quad] / 4$$

$$u = y \quad du = dy$$

$$dv = e^{2y} dy \Rightarrow v = \frac{1}{2} e^{2y}$$

$$\boxed{\frac{y}{2} e^{2y} - e^{2y} = -e^{-x} - \frac{1}{4} e^{-4x} + C} \quad] / 2$$

$$\boxed{2} \quad y dx - 4(x + y^8) dy = 0 \Rightarrow y \frac{dx}{dy} = 4x + 4y^8$$

$$\frac{dx}{dy} - \frac{4}{y} x = 4y^7 \Rightarrow P(y) = -\frac{4}{y} \Rightarrow e^{\int P(y) dy} = e^{-4 \ln y} = \frac{1}{y^4}$$

$$Q(y) = 4y^7$$

$$\Rightarrow \frac{d}{dy} \left(\frac{x}{y^4} \right) = 4y^7 \cdot \frac{1}{y^4} dy \Rightarrow \frac{1}{y^4} x = y^4 + C$$

$$\boxed{x(y) = y^8 + C y^4} \quad] / 2$$

$$x-1) u^h + u' + u = 0$$

$$u(0) = 1, \quad u'(0) = 1$$

$$\sum_{n=2}^{\infty} a_n n(n-1) x^{n-1} = \sum_{n=2}^{\infty} a_n n(n-1) x^{n-1} + \sum_{n=1}^{\infty} a_n x^{n-1} + \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{k=1}^{\infty} a_{k+1} (k+1) k x^k = \sum_{k=0}^{\infty} a_{k+2} (k+1)(k+2) x^k + \sum_{k=0}^{\infty} a_{k+1} (k+1) x^k + \sum_{k=0}^{\infty} a_k x^k = 0$$

$$a_0 + a_1 - 2a_2 + \sum_{k=1}^{\infty} \left[a_{k+1} (k+1)k + a_k - a_{k+2} (k+1)(k+2) \right] x^k = 0$$

$$\begin{cases} 2a_2 = a_0 + a_1 \\ a_2 = \frac{a_0 + a_1}{2} \end{cases}$$

$$a_{k+2} = \frac{a_k}{(k+1)(k+2)} + \frac{a_{k+1}(k+1)}{(k+1)(k+2)}$$

$$k=1: a_3 = \frac{a_1}{6} + a_2 \cdot \frac{2}{3} = \frac{a_1}{6} + \frac{1}{3}a_0 + \frac{1}{3}a_1 = \frac{1}{3}a_0 + \frac{1}{2}a_1$$

$$k=2: a_4 = \frac{a_2}{12} + a_3 \cdot \frac{3}{4} = \frac{a_0}{24} + \frac{a_1}{24} + \frac{3}{4} \left(\frac{1}{3}a_0 + \frac{1}{2}a_1 \right) = \frac{7}{24}a_0 + \frac{10}{24}a_1$$

$$\begin{aligned} k=3: a_5 &= \frac{a_3}{20} + a_4 \cdot \frac{4}{5} = \frac{1}{20} \left(\frac{1}{3}a_0 + \frac{1}{2}a_1 \right) + \frac{4}{5} \left(\frac{7}{24}a_0 + \frac{10}{24}a_1 \right) \\ &= \frac{a_0}{60} + \frac{a_1}{40} + \frac{7}{30}a_0 + \frac{1}{3}a_1 = \frac{15}{60}a_0 + \frac{41}{120}a_1 \end{aligned}$$

$$y(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \dots$$

$$= a_0 + a_1 x + \frac{a_0 x^2}{2} + \frac{a_1 x^3}{2} + \frac{1}{3} a_0 x^3 + \frac{1}{2} a_1 x^3 + \frac{7}{24} a_0 x^4 + \frac{1}{2} a_1 x^4$$

$$= a_0 \left[1 + \frac{1}{2} x^2 + \frac{1}{3} x^3 + \frac{7}{24} x^4 + \frac{1}{4} x^5 + \dots \right] + a_1 \left[x + \frac{1}{2} x^2 + \frac{1}{2} x^3 + \frac{10}{24} x^4 + \frac{1}{2} x^5 + \dots \right]$$

$$y(0) = 1 = a_0$$

$$y'(0) = 2 = a_1$$

$$\therefore y(x) = \left[1 + \frac{1}{2} x^2 + \frac{1}{3} x^3 + \frac{7}{24} x^4 + \frac{1}{4} x^5 + \dots \right] + 2 \left[x + \frac{1}{2} x^2 + \frac{1}{2} x^3 + \frac{10}{24} x^4 + \frac{1}{2} x^5 + \dots \right]$$

✓ 2 .

✓ 4

✓ 10

6] undetermined coef. $y'' - 4y' + 4y = (x-1)e^{2x}$

① Ans $y = e^{mx}$ $m^2 - 4m + 4 = 0$
 $m = \frac{4 \pm \sqrt{16-16}}{2} = 2 \therefore y_c(x) = C_1 e^{2x} + C_2 x e^{2x}$

② NH: $y_p = (Ax^3 + Bx^2)e^{2x}$; $y_p' = (3Ax^2 + 2Bx)e^{2x} + 2(Ax^3 + Bx^2)e^{2x}$
 $= (Ax^3 + Bx^2)e^{2x}$
 $y_p'' = (6Ax + 2B)e^{2x} + 2(3Ax^2 + 2Bx)e^{2x} + 4(Ax^3 + Bx^2)e^{2x} + 2(3Ax^2 + 2Bx)e^{2x}$
 $= (6Ax + 2B)e^{2x} + 2(6Ax^2 + 4Bx)e^{2x} + 4(Ax^3 + Bx^2)e^{2x}$

$\therefore 6Ax^2 + 2Be^{2x} + 12Ax^2e^{2x} + 8Bxe^{2x} + 4Ax^3e^{2x} + 4Bx^2e^{2x} - 12Ax^2e^{2x} - 8Bxe^{2x}$
 $- 8Ax^3e^{2x} - 8Bx^2e^{2x} + 4Ax^3e^{2x} + 4Bx^2e^{2x} = xe^{2x} - 2e^{2x}$

e^{2x}	$2B = -2$	$B = -1$
xe^{2x}	$6A + 8B = 1$	$A = \frac{1}{6}$
x^2e^{2x}	$12A + 4B - 12A - 8B + 4B = 0$	
x^3e^{2x}	$4A - 8A + 4A = 0$	

$\therefore y(x) = C_1 e^{2x} + C_2 x e^{2x} + \left(\frac{1}{6}x - 1\right)x^2 e^{2x}$

$$\left. \begin{aligned} L \frac{di}{dt} + Ri &= E(t), \quad R=0,2 \\ E(t) &= 4 \\ i(0) &= 0 \end{aligned} \right\} L(t) = \begin{cases} 1 - \frac{t}{10}, & 0 \leq t \leq 10 \\ 0, & t > 10 \end{cases}$$

2 operation: $(1 - \frac{t}{10}) \frac{di}{dt} + 0,2 i = 4$ $0 \leq t \leq 10$
 $0,2 i = 4 \Rightarrow i(t) = 20 \text{ A}, t \geq 10$

$$(1 - \frac{t}{10}) \frac{di}{dt} + 0,2 i = 4 \Rightarrow \frac{di}{dt} + \frac{0,2}{(1 - \frac{t}{10})} i = \frac{4}{(1 - \frac{t}{10})}$$

linear: $P_{\text{var}} = \frac{0,2}{(1 - 0,1t)}$ $\int P_{\text{var}} dt = \int \frac{0,2}{1 - 0,1t} dt = \frac{0,2}{-0,1} \ln|1 - 0,1t| = -2 \ln|1 - 0,1t|$
 $\frac{du}{dt} = -0,1 dt$

$$\therefore u = \frac{1}{(1 - 0,1t)^2}$$

$$d\left[\frac{1}{(1 - 0,1t)^2} i\right] = \frac{4}{(1 - 0,1t)} \cdot \frac{1}{(1 - 0,1t)^2} dt = 4 \int \frac{dt}{(1 - 0,1t)^3} = \frac{4}{-0,1} \left[\frac{1}{2} (1 - 0,1t)^{-2} \right] + C$$

$$= \frac{20}{(1 - 0,1t)^2} + C$$

$$\therefore i(t) = 20 + C(1 - 0,1t)^2$$

$$i(0) = 0 \Rightarrow 0 = 20 + C \therefore C = -20$$

$$i(10) = 20 \Rightarrow 20 = 20 + -20(1 - 1)^2 \text{ ok.}$$

$$i(t) = 20 - 20(1 - 0,1t)^2 \quad 0 \leq t \leq 10$$

1/2

Concordia University
 Faculty of Engineering and Computer Science
 Applied Ordinary Differential Equations - ENGR213
 Fall 2022 Final Examination

Instructors: E.Erkman, B.Hersey, A.Kokotov, D.Korotkin, E.Moutuou, A.Paradis, A.Shnirelman,
 B.Yousefzadeh

Special instructions: Do all problems. Show all steps or else no marks will be given. Only Faculty approved calculators are allowed. No other aids are allowed. ALL PROBLEMS CARRY THE SAME WEIGHT

1. Find the general solution of the following equation:

$$x^3 \frac{dx}{dy} = y - x^2 y$$

2. Solve the initial value problem:

$$(y^2 \cos x - 3x^2 y - \ln(x+1))dx + (2y \sin x - x^3 - e^y)dy = 0, \quad y(0) = 1$$

3. Solve the initial value problem for the following Bernoulli equation:

$$x \frac{dy}{dx} + y = x^2 y^2, \quad y(1) = 1$$

4. Find the integrating factor which makes the following equation exact and find the general solution:

$$(y^2 + xy^3)dx + (5y^2 - xy + y^3 \sin y)dy = 0$$

5. A thermometer reading 20°C is placed in an oven preheated to a constant temperature. Through a glass window in the oven door, an observer records that the thermometer read 60°C after 1 minute and 80°C after 2 minutes. How hot is the oven?

6. Find the general solution of the following equation:

$$y'' + 6y' + 9y = e^{-3x} + x$$

by the method of undetermined coefficients.

7. Find all solutions of the differential equation

$$y'' + 2y(y')^3 = 0$$

8. A mass weighting 4 pounds is attached to a spring whose constant is 2 lb/ft. The medium offers a damping force which is proportional to the instantaneous velocity with the coefficient 1 lb/(m/sec). The mass is initially released from a point 1 foot above the equilibrium position with a downward velocity of 8 ft/s. Determine the time at which the mass passes through the equilibrium position. Find the time at which the mass attains its extreme displacement from the equilibrium position. What is the position of the mass at this instant?

(Hint: the acceleration of free fall in Imperial system is assumed to be equal to 32 ft/s^2).

9. Find the general solution of the system

(6)

$$\frac{dx}{dt} = -x + y + t$$

$$\frac{dy}{dt} = 4x - y + e^{4t}$$

by the method of your choice.

10. Use the power series method to find the solution of the initial value problem

(7)

$$y'' - xy' + y = 0, \quad y(0) = 1, \quad y'(0) = -1$$

(write the first four non-zero terms of the power series solution centred at $x = 0$).

Q.1] Find The general solution of the following equation:

$$x^3 \frac{dx}{dy} = y - x^2 y$$

Simplest way: $\Rightarrow x^3 \frac{dx}{dy} = y(1 - x^2)$

$$\frac{x^3 dx}{(1-x^2)} = y dy.$$

(A) (B) $\Rightarrow \int y dy = \frac{y^2}{2} + C$

$$\begin{aligned} \textcircled{A} -x \left(\frac{-x^2}{(1-x^2)} \right) dx &= -x \left(\frac{1-x^2}{(1-x^2)} \right) dx = -x \left(\frac{1-x^2}{(1-x^2)} - \frac{1}{(1-x^2)} \right) dx \\ &= \left(-x + \frac{x}{(1-x^2)} \right) dx \end{aligned}$$

$$\therefore \int \left(-x + \frac{x}{(1-x^2)} \right) dx = -\frac{x^2}{2} + \int \frac{x}{(1-x^2)} dx = -\frac{x^2}{2} + \frac{1}{2} (-1) \ln|1-x^2|$$

$u=1-x^2$
 $du=-2x dx$

$$\therefore \textcircled{A} = \textcircled{B}$$

$$\begin{aligned} \Rightarrow \frac{1}{2} (x^2 + \ln|1-x^2|) &= \frac{y^2}{2} + C \\ -x^2 - \ln|1-x^2| &= y^2 + C_1 \end{aligned}$$

~~the~~

1- Solve the initial value problem

$$(y^2 \cos x - 3x^2 y - \ln(x+1)) dx + (2y \sin x - x^3 - e^y) dy = 0, y(0) = 1$$

by the form, check if exact

$$\begin{aligned} M &= y^2 \cos x - 3x^2 y - \ln(x+1) \rightarrow \frac{\partial M}{\partial y} = 2y \cos x - 3x^2 \\ N &= 2y \sin x - x^3 - e^y \rightarrow \frac{\partial N}{\partial x} = 2y \cos x - 3x^2 \end{aligned} \quad \left. \vphantom{\begin{aligned} M &= y^2 \cos x - 3x^2 y - \ln(x+1) \\ N &= 2y \sin x - x^3 - e^y \end{aligned}} \right\} \text{OK.}$$

$$\frac{\partial f(x,y)}{\partial x} = M \quad \text{or} \quad \frac{\partial f}{\partial y} = N$$

$$f(x,y) = \int N dy + g(x) \rightarrow \int 2y \sin x dy - \int x^3 dy - \int e^y dy + g(x)$$

$$f(x,y) = y^2 \sin x - x^3 y - e^y + g(x)$$

$$\text{Now, } \frac{\partial f}{\partial x} = y^2 \cos x - 3x^2 y + g'(x) = y^2 \cos x - 3x^2 y - \ln(x+1)$$

$$\therefore g'(x) = -\ln(x+1) \rightarrow \text{not easy: } \int \ln u du = u \ln u - u$$

$$g(x) = -[(x+1)\ln(x+1) - (x+1)]$$

$$\begin{aligned} u &= \ln u \quad du = \frac{1}{u} du \\ du &= \frac{1}{u} du \rightarrow u = 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} u &= \ln u \\ du &= \frac{1}{u} du \end{aligned}} \right\} \begin{aligned} u \ln u &= \int \frac{u}{u} du \\ u \ln u &= \int 1 du \\ u \ln u &= u + C \end{aligned}$$

$$\Rightarrow f(x,y) = y^2 \sin x - x^3 y - e^y - (x+1)\ln(x+1) + (x+1) = C$$

$$y(0) = 1 \rightarrow 1^2 \sin(0) - 0 - e^1 - \ln(1) + 1 = C \rightarrow C = 1 - e$$

$$f(x,y) = y^2 \sin x - x^3 y - e^y - (x+1)\ln(x+1) + x = -e$$

1. Solve the initial value problem for the following Bernoulli equation:

$$x \frac{dy}{dx} + y = x^2 y^3, \quad y(1) = 1$$

$$n=2, \quad y=u^{-1} \quad \left(\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = -\frac{1}{u^2} \frac{du}{dx} \right)$$

$$\therefore -\frac{x}{u^2} \frac{du}{dx} + \frac{1}{u} = \frac{x^2}{u^3} \Rightarrow \frac{du}{dx} - \frac{1}{x} u = -x$$

Integrating factor: $P(x) = -\frac{1}{x} \Rightarrow \int P(x) dx = -\int \frac{1}{x} dx = -\ln x = \frac{1}{x}$

$$\therefore \frac{d}{dx} \left(\frac{1}{x} u \right) = -1 \Rightarrow \frac{1}{x} u = -x + C$$

$$u = -x^2 + Cx \Rightarrow y = \frac{1}{Cx - x^2}$$

$$y(1) = 1 \Rightarrow 1 = \frac{1}{C-1} \Rightarrow C-1 = 1 \Rightarrow C = 2$$

$$y(x) = \frac{1}{(2x - x^2)}$$

The integrating factor is

and find the general solution:

$$(y^2 + xy^3)dx + (5y^2 - xy + y^3 \sin(y))dy = 0$$

$$\text{let } P = y^2 + xy^3 \quad ; \quad Q = 5y^2 - xy + y^3 \sin(y) \quad \mu(y) = \exp\left[\int \left(\frac{\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}}{P}\right) dy\right]$$

$$\frac{\partial P}{\partial y} = 2y + 3xy^2 \quad ; \quad \frac{\partial Q}{\partial x} = -y$$

$$\textcircled{1} \frac{2y + 3xy^2 + y}{5y^2 - xy + y^3 \sin(y)}$$

$$\textcircled{1} \frac{\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}}{Q} (x) \quad \text{or} \quad \textcircled{2} \frac{\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}}{P} (y)$$

$$\textcircled{2} \frac{-y - 2y - 3xy^2}{y^2 + xy^3} = \frac{-3y[1 + xy]}{y^2[1 + xy]} = -\frac{3}{y} \quad \text{ok!}$$

$$\therefore \mu(y) = \int \frac{3}{y} dy \quad e^{\int -\frac{3}{y} dy} = e^{-3 \ln y} = \frac{1}{y^3} \quad \square$$

$$\therefore \mu \left(\left(\frac{1}{y} + x \right) dx + \left(\frac{5}{y} - \frac{x}{y} + \sin(y) \right) dy \right) = 0 \quad \underline{\underline{\text{OK}}}$$

$$\text{let } f = \int \left(\frac{1}{y} + x \right) dx + g(y)$$

$$= \frac{x}{y} + \frac{x^2}{2} + g(y)$$

$$\frac{\partial f}{\partial y} = -\frac{x}{y^2} + g'(y) = \frac{5}{y} - \frac{x}{y} + \sin(y) \quad \therefore g'(y) = \frac{5}{y} + \sin(y)$$

$$g(y) = 5 \ln|y| - \cos(y) + C_1$$

$$\Rightarrow f(C_1) = C \Rightarrow \boxed{\frac{x}{y} + \frac{x^2}{2} + 5 \ln|y| - \cos(y) = C}$$

#5 Newton's law of cooling $\rightarrow \frac{dT}{dt} = k(T - T_{\text{amb}})$

$T_{\text{amb}} = 20^\circ\text{C} \rightarrow t=0$ / $T_a = 60^\circ\text{C}$, $t=1\text{w}$ / $T = 80^\circ\text{C}$ $t=2\text{w}$

over: T_{amb} .

$$\Rightarrow \frac{dT}{dt} = k(T - T_{\text{amb}})$$

$$\frac{dT}{T - T_{\text{amb}}} = k dt \rightarrow \ln|T - T_{\text{amb}}| = kt + C$$

$$T = T_{\text{amb}} + C_1 e^{kt}$$

condition

$$\left. \begin{aligned} 20 &= T_{\text{amb}} + C_1 \\ 60 &= T_{\text{amb}} + C_1 e^k \\ 80 &= T_{\text{amb}} + C_1 e^{2k} \end{aligned} \right\} 3 \text{ unknown.}$$

$$C_1 = 20 - T_{\text{amb}} \rightarrow \begin{aligned} 60 &= T_{\text{amb}} + (20 - T_{\text{amb}}) e^k \rightarrow \frac{60 - T_{\text{amb}}}{20 - T_{\text{amb}}} = e^k \\ 80 &= T_{\text{amb}} + (20 - T_{\text{amb}}) e^{2k} \end{aligned}$$

$$\therefore 80 = T_{\text{amb}} + (20 - T_{\text{amb}}) \frac{(60 - T_{\text{amb}})^2}{(20 - T_{\text{amb}})^2}$$

$$(80 - T_a)(20 - T_a) = (60 - T_a)^2$$

$$1600 - 100T_a + T_a^2 = 3600 - 120T_a + T_a^2$$

$$20T_a = 2000$$

$$\boxed{T_a = 100^\circ\text{C}}$$

Find the general solution by undetermined coeff.

$$y'' + 6y' + 9y = e^{-3x} + x$$

① Homog: $y'' + 6y' + 9y = 0$

try $e^{mx} \Rightarrow m^2 + 6m + 9 = 0$

$$m = \frac{-6 \pm \sqrt{36 - 36}}{2} = -3$$

$$\therefore y(x) = C_1 e^{-3x} + C_2 x e^{-3x}$$

② P.N.H: $f(x) = e^{-3x} + x \Rightarrow y_p = A x^2 e^{-3x} + B x + C$

$$A e^{-3x} (x^2)$$

$$\Rightarrow y_p' = 2A x e^{-3x} + A x^2 (-3) e^{-3x} + B$$

$$y_p'' = 2A e^{-3x} - 6A x e^{-3x} - 6A x e^{-3x} + 9A x^2 e^{-3x}$$

$$\therefore y'' + 6y' + 9y = e^{-3x} + x$$

$$2A e^{-3x} - 12A x e^{-3x} + 9A x^2 e^{-3x} + 12A x e^{-3x} - 18A x^2 e^{-3x} + 6B e^{-3x} + 9A x^2 e^{-3x} + 9B x + 9C = e^{-3x} + x$$

$$\left. \begin{array}{l} x^0: 6B + 9C = 0 \\ x^1: 9B = 1 \\ e^{-3x}: 2A = 1 \end{array} \right\} \begin{array}{l} A = \frac{1}{2} \\ B = \frac{1}{9} \\ C = -\frac{6}{81} \end{array} \Rightarrow y(x) = C_1 e^{-3x} + C_2 x e^{-3x} + \frac{x^2}{2} e^{-3x} + \frac{1}{9} x - \frac{6}{81}$$

$$① \quad y'' + 2y(y')^3 = 0 \quad \leadsto F(y, y', y'') = 0$$

$$\therefore u = y' ; \quad y'' = u \frac{du}{dy}$$

$$u \frac{du}{dy} + 2y u^3 = 0 \quad \leadsto \frac{u du}{dy} = -2y u^3$$

$$\boxed{\frac{du}{u^2} = -2y dy}$$

$$\rightarrow -\frac{1}{u} = -\frac{2y^2}{2} + C \quad \leadsto \frac{du}{dx} = \frac{1}{y^2 + C}$$

$$(y^2 + C) dy = dx$$

$$\boxed{\frac{y^3}{3} + Cy = x + C_1}$$

9) find the general solution.

$$\frac{dx}{dt} = -x + y + t$$

$$\frac{dy}{dt} = 4x - y + e^{4t}$$

$$\Rightarrow \dot{X} = \begin{pmatrix} -1 & 1 \\ 4 & -1 \end{pmatrix} X + \begin{pmatrix} t \\ e^{4t} \end{pmatrix}$$

① Char: $\Rightarrow \begin{vmatrix} -1-\lambda & 1 \\ 4 & -1-\lambda \end{vmatrix} = 0 \Rightarrow 1 + 2\lambda + \lambda^2 - 4 = 0$
 $\lambda^2 + 2\lambda - 3 = 0$
 $\lambda = \frac{-2 \pm \sqrt{4+12}}{2} = \begin{Bmatrix} 1 \\ -3 \end{Bmatrix}$

$\therefore \lambda_1 = 1 \Rightarrow \begin{pmatrix} -2 & 1 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} k_{11} \\ k_{12} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow -2k_1 + k_2 = 0 \Rightarrow \frac{k_2}{k_1} = 2 \rightarrow K_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$\lambda_2 = -3 \Rightarrow \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} k_{21} \\ k_{22} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow 2k_1 = -k_2 \Rightarrow K_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$

$\therefore X_c = C_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^t + C_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-3t}$

② N-H: let us try $X_p = \begin{pmatrix} A + B + Ce^{4t} \\ D + E + Fe^{4t} \end{pmatrix} \Rightarrow \dot{X}_p = \begin{pmatrix} A + 4Ce^{4t} \\ D + 4Fe^{4t} \end{pmatrix}$

$\therefore \begin{pmatrix} A + 4Ce^{4t} \\ D + 4Fe^{4t} \end{pmatrix} = \begin{pmatrix} -A + B + C + D + E + Fe^{4t} + t \\ 4A + 4B + 4C + D + E + Fe^{4t} + e^{4t} \end{pmatrix}$ ①

① $t \sim A = -B + E$ ③
 $t^1 \sim 0 = -A + B + 1$ ④
 $e^{4t} \sim 4C = -C + F$ ⑤

② $t^0 \sim D = 4B - E$ ⑥
 $t^1 \sim 0 = 4A - D$ ⑦
 $e^{4t} \sim 4F = 4C - F + 1$ ⑧

$$\rightarrow D = 4A \quad \text{in (6)} \quad -1 = -A + 4A = 3A$$

$$A = -\frac{1}{3}, \quad D = -\frac{4}{3}$$

$$\textcircled{c} \rightarrow 5C = F$$

$$\textcircled{d} \rightarrow 5F = 4C + 1 \rightarrow 5F - 4C = 1 \quad \text{or} \quad 5C = 1 \quad \left[C = \frac{1}{5}; \quad F = \frac{5}{5} = 1 \right]$$

$$\textcircled{a} \rightarrow -\frac{1}{3} = -B + E \rightarrow B = E + \frac{1}{3}$$

$$\textcircled{b} \rightarrow -\frac{4}{3} = 4E + \frac{1}{3} - E \rightarrow -\frac{8}{3} = 3E \rightarrow E = -\frac{8}{9} \quad \& \quad B = -\frac{5}{9}$$

$$\rightarrow X(t) = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{-3t} + \begin{pmatrix} -\frac{1}{3}t - \frac{5}{9} + \frac{1}{5}e^{4t} \\ -\frac{4}{3}t - \frac{8}{9} + \frac{5}{5}e^{4t} \end{pmatrix}$$

□

$$x(t) = C_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^t + C_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-3t} + \begin{pmatrix} -1 - \frac{1}{3}t + (\frac{1}{12} - \frac{1}{24})e^{4t} \\ -1 - \frac{1}{4}t + (\frac{1}{12} - \frac{1}{24})e^{4t} \end{pmatrix}$$

#10) problem: $y'' - xy' + y = 0$ $y(0) = 1$ $y'(0) = -1$. 4 marks.

$$\Rightarrow y = \sum_{n=0}^{\infty} a_n x^n ; y' = \sum_{n=1}^{\infty} n a_n x^{n-1} ; y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

$$\Rightarrow \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - \sum_{n=1}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} a_n x^n = 0$$

$k=n-2 \quad n=k+2 \quad k=n$

$$\sum_{k=0}^{\infty} (k+2)(k+1) a_{k+2} x^k - \sum_{k=1}^{\infty} k a_k x^k + \sum_{k=0}^{\infty} a_k x^k = 0$$

$$2a_2 + a_0 + \sum_{k=1}^{\infty} [(k+2)(k+1) a_{k+2} - k a_k + a_k] x^k = 0$$

$$\Rightarrow 2a_2 + a_0 = 0 \Rightarrow a_2 = -\frac{1}{2} a_0, a_3 = 0$$

$$a_{k+2} = \frac{(1-k) a_k}{(k+2)(k+1)}$$

$k=1$	$a_3 = 0$
$k=2$	$a_4 = \frac{-a_2}{4 \cdot 3} = \frac{1}{2} \frac{a_0}{12}$
$k=3$	$a_5 = \frac{-2a_3}{5 \cdot 4} = 0$
$k=4$	$a_6 = \frac{-3a_4}{6 \cdot 5}$

$$\therefore y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$$

$$= a_0 + a_1 x + \left(-\frac{1}{2} a_0\right) x^2 + \frac{1}{2} \frac{a_0}{12} x^4 + \dots$$

$$= a_0 \left[1 - \frac{1}{2} x^2 + \frac{1}{24} x^4 + \dots \right] + a_1 x$$

$$y(0) = 1 = a_0 \quad y'(0) = -1 = a_1 \quad \therefore y(x) = \left[1 - \frac{1}{2} x^2 + \frac{1}{24} x^4 + \dots \right] - x$$

Mock exam

Summary

Quiz

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82

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